

Generalized Quantifiers

Week 5

Slides and materials based on the courses by
Noortje Venhuizen and Mareike Hartmann

Back to Noun Phrases

- Natural language contains a wide variety of NPs, serving as **quantifiers**:
 - *all students, no woman, not every man, everything, nothing, three books, the ten professors, John, John and Mary, only John, firemen, at least five horses, most girls, all but ten marbles, less than half of the audience, John's car, some student's exercise, no student except Mary, more male than female cats, each other*



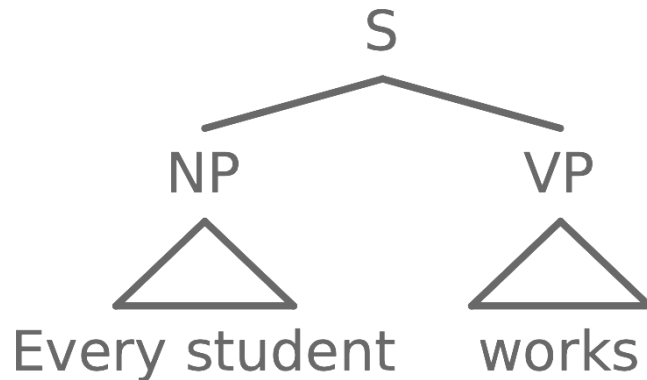
Aristotle: “all quantifiers are second-order relations between sets”



Frege: “all quantifiers can be defined in terms of logical quantifiers ($\forall$, $\exists$)”

NP interpretation

“every student” $\mapsto \lambda P \forall x(\text{student}(x) \rightarrow P(x))$



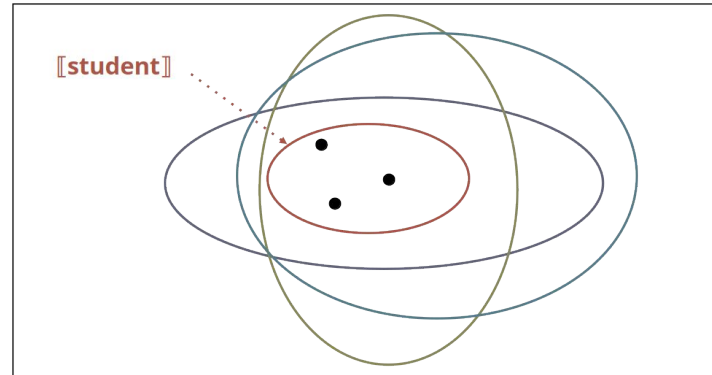
- $\llbracket \text{every student} \rrbracket \in D_{\langle\langle e, t \rangle, t \rangle}$
- $D_{\langle\langle e, t \rangle, t \rangle}$ is the set of functions from properties to truth values
 - In other words: “every student” denotes the set of properties that apply to every student (property = set of individuals)
 - $\llbracket \text{Every student} \rrbracket^M = \{ P \subseteq U_M \mid \text{every student has property } P \}$
 $= \{ P \subseteq U_M \mid \llbracket \text{student} \rrbracket \subseteq P \}$
 - $\llbracket \text{every student works} \rrbracket^M = 1$ iff $\llbracket \text{work} \rrbracket^M \in \llbracket \text{every student} \rrbracket^M$

Generalised quantifiers

- Generalised quantifiers are sets of subsets of U_M
 - i.e. sets of properties
 - i.e. elements of $D_{\langle\langle e, t \rangle, t \rangle}$
- “*every student*” $\mapsto \lambda P \forall x(\text{student}(x) \rightarrow P(x))$
 - $\llbracket \text{every student} \rrbracket^M = \{ P \subseteq U_M \mid \llbracket \text{student} \rrbracket^M \subseteq P \}$
- “*a student*” $\mapsto \lambda P \exists x(\text{student}'(x) \wedge P(x))$
 - $\llbracket \text{a student} \rrbracket^M = \{ P \subseteq U_M \mid \llbracket \text{student} \rrbracket^M \cap P \neq \emptyset \}$
- “*Bill*” $\mapsto \lambda P.P(\text{bill}^*)$
 - $\llbracket \text{bill} \rrbracket^M = \{ P \subseteq U_M \mid \llbracket \text{bill}^* \rrbracket^M \in P \}$

Universal quantifiers: denotation

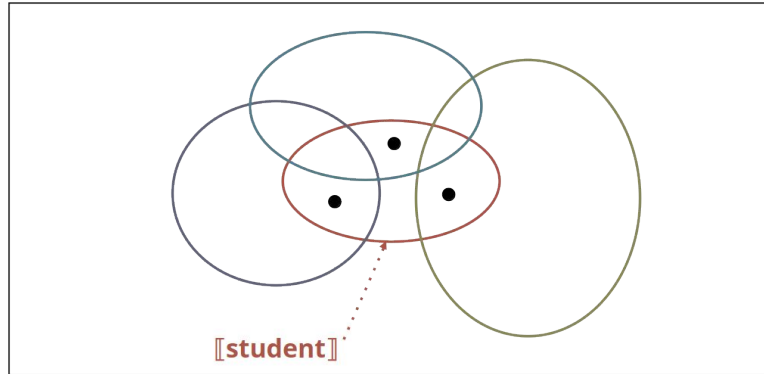
- “*every student*” denotes the set of properties that apply to every student
 - i.e., all supersets of $\llbracket \text{student} \rrbracket$



- $\llbracket \text{every } N \rrbracket^M = \{ P \subseteq U_M \mid \llbracket N \rrbracket^M \subseteq P \}$

Existential quantifiers: denotation

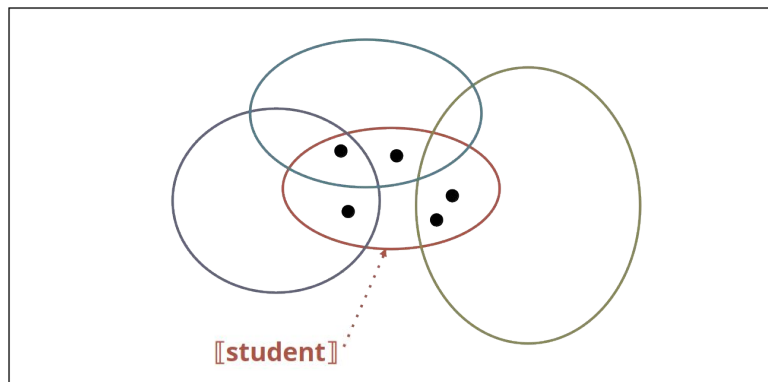
- “*a student*” denotes the set of properties that apply to at least one student
 - i.e., all sets that intersect with $\llbracket \textit{student} \rrbracket$



- $\llbracket a(n) N \rrbracket^M = \{ P \subseteq U_M \mid \llbracket N \rrbracket^M \cap P \neq \emptyset \}$

Cardinal quantifiers: denotation

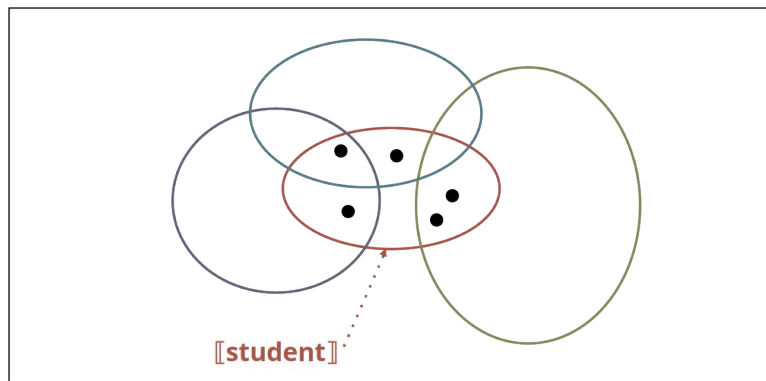
- “two students” denotes the set of properties that apply to at least two students
 - $\{ P \subseteq U_M \mid |\llbracket student \rrbracket^M \cap P| \geq 2 \}$



- $\llbracket K N \rrbracket^M = \{ P \subseteq U_M \mid |\llbracket N \rrbracket^M \cap P| \geq K \}$ (probable exception: zero)

Named entities as (generalized) quantifiers

- “*Bill*” denotes the set of properties that apply to Bill
 - $\llbracket bill \rrbracket^M = \{ P \subseteq U_M \mid \llbracket bill^* \rrbracket^M \in P \} = \{ P \subseteq U_M \mid \{ \llbracket bill^* \rrbracket^M \} \subseteq P \}$



- $\llbracket N_{named} \rrbracket^M = \{ P \subseteq U_M \mid \llbracket N_{named}^* \rrbracket^M \in P \}$

Exercise

1. $\llbracket \textit{not all } N \rrbracket^M$
2. $\llbracket \textit{no } N \rrbracket^M$
3. $\llbracket \textit{half of the } N \rrbracket^M$
4. $\llbracket \textit{most of the } N \rrbracket^M$
5. $\llbracket \textit{all but ten of the } N \rrbracket^M$

Generalised Quantifier Theory: central questions

1. How do generalised quantifiers *differ* in terms of their formal properties?
2. What *universal* regularities govern the meaning of terms?
3. Which *subclasses* represent meanings of natural language noun phrases?

Observation 1: inference patterns

- **all men** walked rapidly $\models$ **all men** walked
 - **all men** walked $\not\models$ **all men** walked rapidly
 - **no man** walked $\models$ **no man** walked rapidly
 - **no man** walked rapidly $\not\models$ **no man** walked
-
- **a girl** smoked a cigar $\models$ **a girl** smoked
 - **a girl** smoked $\not\models$ **a girl** smoked a cigar
 - **few girls** smoked $\models$ **few girls** smoked a cigar
 - **few girls** smoked a cigar $\not\models$ **few girls** smoked

Observation 2: negative polarity items

- **Negative polarity items** (NPIs: “*at all*”, “*anymore*”, etc.) typically occur only in contexts with negation:
 - “*she doesn’t like driving at all*” vs.
 - **“*she likes driving at all*”
- What formally licenses Negative Polarity Items?
 - “*nobody saw anything*” vs. **“*somebody saw anything*”
 - “*no student has ever been to Saarbrücken*”
vs. **“*a student has ever been to Saarbrücken*”
 - But: “*few students have ever been to Saarbrücken*”

Observation 3: coordination

- *“no man and few women walked”*
- *“none of the girls and at most three boys walked”*
- ***“some man and few women walked”*
- ***“John and no woman saw Jane”*

Explaining Observation 1: subsets and supersets

- “all men walked rapidly” $\models$ “all men walked”
 - $\llbracket \textit{to walk rapidly} \rrbracket \subseteq \llbracket \textit{to walk} \rrbracket$
- “a girl smoked a cigar” $\models$ “a girl smoked”
 - $\llbracket \textit{to smoke a cigar} \rrbracket \subseteq \llbracket \textit{to smoke} \rrbracket$
- Intuitively: For the given quantifiers, the sentence $[_S \text{ NP VP}]$ remains true if the denotation of the VP is made “larger”

Upward monotonicity

- A quantifier Q is (*right*) **upward monotonic** (or: monotone increasing) in $M = (U, V)$ iff Q is “closed under supersets”, i.e.:
 - for all $X, Y \subseteq U$: if $X \in Q$ and $X \subseteq Y$, then $Y \in Q$
- A noun phrase is (right) upward monotonic if it denotes a (right) upward monotonic quantifier

Upward monotonicity: entailment tests

- If $[[VP_1]] \subseteq [[VP_2]]$, then $NP VP_1 \models NP VP_2$
 - $[[to\ walk\ rapidly]] \subseteq [[to\ walk]]$
 - “all men walked rapidly” $\models$ “all men walked”
 - “no man walked rapidly” $\not\models$ “no man walked”
- $NP (VP_1\ and\ VP_2) \models (NP VP_1)\ and\ (NP VP_2)$
where: $[[VP1\ and\ VP2]] = [[VP1]] \cap [[VP2]]$
 - “all men smoked and drank” $\models$ “all men smoked and all men drank”
 - “no man smoked and drank” $\not\models$ “no man smoked and no man drank”

Downward monotonicity

- A quantifier Q is (*right*) **downward monotonic** (or: monotone *decreasing*) in $M = (U, V)$ iff Q is “closed under subsets”, i.e.:
 - for all $X, Y \subseteq U$: if $X \in Q$ and $Y \subseteq X$, then $Y \in Q$
- “*no man walked*” $\models$ “*no man walked rapidly*”
 - $\llbracket \text{to walk rapidly} \rrbracket \subseteq \llbracket \text{to walk} \rrbracket$
- “*few girls smoked*” $\models$ “*few girls smoked a cigar*”
 - $\llbracket \text{to smoke a cigar} \rrbracket \subseteq \llbracket \text{to smoke} \rrbracket$
- A noun phrase is downward monotonic if it denotes a downward monotonic quantifier

Downward monotonicity: entailment tests

- If $\llbracket \text{VP}_2 \rrbracket \subseteq \llbracket \text{VP}_1 \rrbracket$, then $\text{NP VP}_1 \models \text{NP VP}_2$
 - $\llbracket \text{to walk rapidly} \rrbracket \subseteq \llbracket \text{to walk} \rrbracket$
 - “no man walked” $\models$ “no man walked rapidly”
 - “all men walked” $\not\models$ “all men walked rapidly”
- $\text{NP (VP}_1 \text{ or VP}_2) \models (\text{NP VP}_1) \text{ and } (\text{NP VP}_2)$
where: $\llbracket \text{VP}_1 \text{ or VP}_2 \rrbracket = \llbracket \text{VP}_1 \rrbracket \cup \llbracket \text{VP}_2 \rrbracket$
and $\llbracket \text{VP}_1 \text{ and VP}_2 \rrbracket = \llbracket \text{VP}_1 \rrbracket \cap \llbracket \text{VP}_2 \rrbracket$
 - “neither girl drank or smoked” $\models$ “neither girl drank and neither girl smoked”
 - “all boys sing or dance” $\not\models$ “all boys sing and all boys dance”

Explaining Observation 2

- “*nobody saw anything*” vs. **“*somebody saw anything*”
- “*no student has ever been to Saarbrücken*”
vs. **“*a student has ever been to Saarbrücken*”
 - But: “*few students have ever been to Saarbrücken*”
- NPIs are licensed only in (right) downward monotonic contexts

Explaining Observation 3

- *“no man and few women walked”*
- *“none of the girls and at most three boys walked”*
- ***“some man and few women walked”*
- ***“John and no woman saw Jane”*
- NPs can only be coordinated when they have the same direction of (right) monotonicity

Monotonicity and logical operators

- (Right) monotonic quantifiers are ***closed under*** conjunction and disjunction:
 - “all boys and a girl walked rapidly” $\models$ “all boys and a girl walked”
 - “John or a student arrived late” $\models$ “John or a student arrived”
 - where:
$$\begin{aligned} \llbracket \text{NP}_1 \text{ and } \text{NP}_2 \rrbracket &= \llbracket \text{NP}_1 \rrbracket \cap \llbracket \text{NP}_2 \rrbracket \\ \llbracket \text{NP}_1 \text{ or } \text{NP}_2 \rrbracket &= \llbracket \text{NP}_1 \rrbracket \cup \llbracket \text{NP}_2 \rrbracket \end{aligned}$$
- The intersection/union of two monotonic quantifiers is a quantifier with the same direction of monotonicity

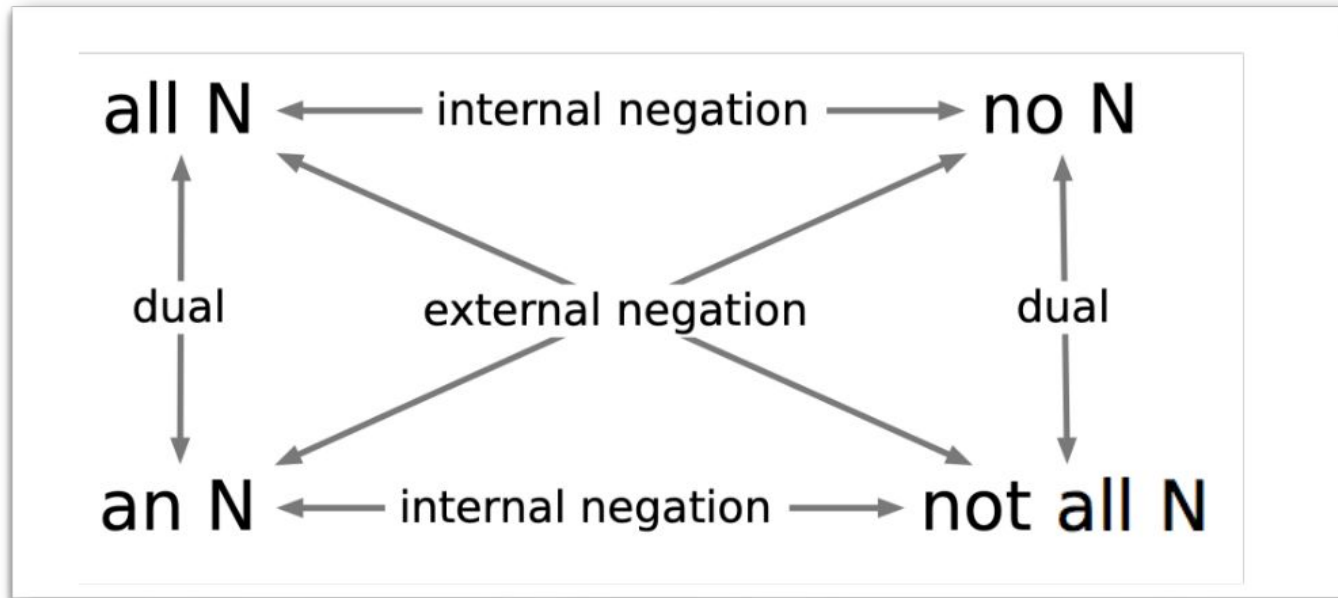
Monotonicity and negation

- External negation: $\neg Q = \{ P \subseteq U_M \mid P \notin Q \}$
 - $\llbracket \text{not all } N \rrbracket = \{ P \subseteq U_M \mid P \notin \llbracket \text{all } N \rrbracket \}$
 $= \{ P \subseteq U_M \mid \llbracket N \rrbracket \not\subseteq P \} = \neg \llbracket \text{all } N \rrbracket$
- Internal negation: $\lambda R. Q(\neg R) = \{ P \subseteq U_M \mid (U_M - P) \in Q \}$
 - $\llbracket \text{all } N \text{ don't} \rrbracket = \{ P \subseteq U_M \mid \llbracket N \rrbracket \subseteq U_M - P \}$
 $= \{ P \subseteq U_M \mid \llbracket N \rrbracket \cap P = \emptyset \} = \llbracket \text{no } N \rrbracket$
- Internal and external negation of a quantifier both flip the direction of monotonicity (*upward* $\Rightarrow$ *downward* and *downward* $\Rightarrow$ *upward*)

Duals

- The **dual** Q^* of a quantifier Q in M is defined as the external *and* internal negation of Q : $Q^* = \lambda R. \neg Q(\neg R)$
 - Each quantifier is its own “double dual”: $(Q^*)^* = Q$
 - $\llbracket Q^* \rrbracket = \{ P \subseteq U_M \mid U_M - P \notin Q \}$
 - $\forall = \exists^*$ (which means?)
- The dual preserves (right) monotonicity:
 - If Q is upward monotonic, then Q^* is upward monotonic
 - If Q is downward monotonic, then Q^* is downward monotonic

The “Square of Opposition”



The myth of Language importance

The universal basis of linguistic exceptionalism

doi:10.1017/S0140525X09991130

Daniel Harbour

Department of Psychology
University of Cambridge
harbour@psychology.cam.ac.uk
<http://www.psychology.cam.ac.uk/people/dan.harbour/>

The myth of linguistic universals

doi:10.1017/S0140525X09990744

Morten H. Christiansen

^aDepartment of Psychology, Santa Fe Institute, Santa Fe, NM 87501, USA
christiansen@cornell.edu
<http://www.psych.cornell.edu/people/Faculty/mhc27.htm>
n.chater@ucl.ac.uk
http://www.psychol.ucl.ac.uk/people/profiles/chater_nick.htm

Universal grammar is dead

doi:10.1017/S0140525X09990744

Michael Tomasello

Max Planck Institute for Evolutionary Anthropology, D-04103 Leipzig, Germany.
tomas@eva.mpg.de

Abstract: The idea of a biologically evolved, universal grammar with linguistic content is a myth, perpetuated by three spurious explanatory strategies of generative linguists. To make progress in understanding human linguistic competence, cognitive scientists must abandon the idea of an innate universal grammar and instead try to build theories that explain both linguistic universals and diversity and how they emerge.

Universal grammar is, and has been for some time, a completely empty concept. Ask yourself: what exactly is in universal grammar? Oh, you don't know – but you are sure that the experts (generative linguists) do. Wrong: they don't. And not only that, they have no method for finding out. If there is a method, it would be looking carefully at all the world's thousands of languages to discern universals. But that is what linguistic typologists have been doing for the past several decades, and, as Evans & Levinson (E&L) report, they find no universal grammar.

^bSanta Fe Institute, Santa Fe, NM 87501.
pinker@wjh.harvard.edu
<http://pinker.wjh.harvard.edu>
Ray.Jackendoff@tufts.edu
<http://ase.tufts.edu/cogstud/incbios/RayJackendoff/index.htm>

Linguistic universals: Abstract but not

doi:10.1017/S0140525X09990604

David M. Reiss, Rutgers University, New Brunswick, NJ 08901.
reiss@psychology.rutgers.edu
reiss@psychology.rutgers.edu/~mabaker/

Linguistic universals: Abstract but not

doi:10.1017/S0140525X09990604

Ray Jackendoff^{b,c}
University, Cambridge, MA 02138;
University, Medford, MA 02155; and

Looking for Universals I: monotonicity constraint

“The simple noun phrases of any natural language express monotone quantifiers or conjunctions of monotone quantifiers.”

(Barwise & Cooper, 1981)

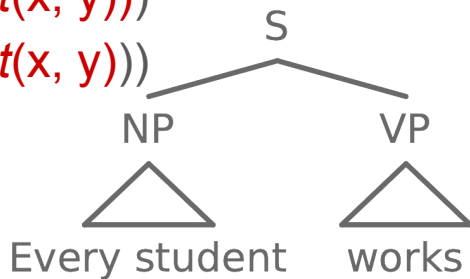
- **Simple noun phrase:** Proper names or NPs of the form $[\text{DET } N]_{\text{NP}}$
- **Monotone quantifiers:** quantifiers that are (right) upward or downward monotonic

From NPs to determiners

- “*every man walked*” $\mapsto \forall x(\text{man}(x) \rightarrow \text{walk}(x))$
 - $\llbracket \text{every} \rrbracket = \llbracket \lambda P \lambda Q \forall x (P(x) \rightarrow Q(x)) \rrbracket \in D_{\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle}$
 - $\llbracket \text{every} \rrbracket(A)(B) = 1$ iff $A \subseteq B$
- *Syntactically*, determiners are expressions that take a noun and a verb phrase to form a sentence
- *Semantically* the interpretation of a determiner can be seen as:
 - a function from sets of entities to sets of properties: $\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$
 - a relation between two sets A and B, denoted by the NP and VP, respectively

Quick aside: restriction and nuclear scope

- Logic: **scope** of a *quantifier*
 - $\underline{\forall x}((\text{person}(x) \wedge \text{hungry}(x)) \rightarrow \exists y(\text{banana}(y) \wedge \text{eat}(x, y)))$
 - $\forall x((\text{person}(x) \wedge \text{hungry}(x)) \rightarrow \underline{\exists y}(\text{banana}(y) \wedge \text{eat}(x, y)))$
- Linguistics: **restriction** (denotation of the noun the determiner combines with to form an NP) and **nuclear scope** (**scope** - **restriction**) of a *determiner*
 - $\underline{\forall x}((\text{person}(x) \wedge \text{hungry}(x)) \rightarrow \exists y(\text{banana}(y) \wedge \text{eat}(x, y)))$
 - $\forall x((\text{person}(x) \wedge \text{hungry}(x)) \rightarrow \underline{\exists y}(\text{banana}(y) \wedge \text{eat}(x, y)))$



A note on monotonicity

- We've been talking about *right* monotonicity (and will be for the remainder of the course): *right* monotonicity $\Rightarrow$ nuclear scope
 - *left* monotonicity $\Rightarrow$ *restriction*

	Decreasing	Increasing
Right	$\neg \exists:$ “no dog <u>moved</u> ” $\models$ “no dog <u>walked</u> ” $\neg \forall:$ “not all dogs <u>moved</u> ” $\models$ “not all dogs <u>walked</u> ”	$\exists:$ “a dog walked” $\models$ “a dog <u>moved</u> ” $\forall:$ “all dogs walked” $\models$ “all dogs <u>moved</u> ”
Left	$\neg \exists:$ “no <u>person</u> walked” $\models$ “no <u>man</u> walked” $\forall:$ “all <u>people</u> walked” $\models$ “all <u>men</u> walked”	$\exists:$ “a <u>woman</u> walked” $\models$ “a <u>person</u> walked” $\neg \forall:$ “not all <u>women</u> walked” $\models$ “not all <u>people</u> walked”

Persistence (left upward monotonicity)

- A determiner D is **persistent** in M iff: for all X, Y, Z :
 - if $D(X, Z)$ and $X \subseteq Y$, then $D(Y, Z)$
- Persistence test: If $\llbracket N_1 \rrbracket \subseteq \llbracket N_2 \rrbracket$, then $\text{DET } N_1 \text{ VP} \models \text{DET } N_2 \text{ VP}$
 - $\llbracket \text{woman} \rrbracket \subseteq \llbracket \text{person} \rrbracket$
 - “*some women walked*” $\models$ “*some people walked*”
 - $\llbracket \text{boy} \rrbracket \subseteq \llbracket \text{child} \rrbracket$
 - “*at least four boys were smoking*”
 $\models$ “*at least four children were smoking*”

Antipersistence (left downward monotonicity)

- A determiner D is **antipersistent** in M iff: for all X, Y, Z :
 - if $D(X, Z)$ and $Y \subseteq X$, then $D(Y, Z)$
- Antipersistence test: If $\llbracket N_2 \rrbracket \subseteq \llbracket N_1 \rrbracket$, then $\text{DET } N_1 \text{ VP} \models \text{DET } N_2 \text{ VP}$
 - $\llbracket \text{toddler} \rrbracket \subseteq \llbracket \text{children} \rrbracket$
 - “*all children walked*” $\models$ “*all toddlers walked*”
 - $\llbracket \text{girl} \rrbracket \subseteq \llbracket \text{female} \rrbracket$
 - “*no female was smoking*” $\models$ “*no girl was smoking*”

Looking for Universals II: conservativity constraint

“In every natural language, simple determiners together with an N yield an NP which ‘lives on $[[N]]$ ’.” (Barwise & Cooper, 1981)

- A determiner D is **conservative** iff “ D lives on A ”:
for every $A, B \subseteq U$: $D(A, B) \Leftrightarrow D(A, A \cap B)$
 - “all students work” $\Leftrightarrow$ “all students are students that work”
 - “some girls are dancing” $\Leftrightarrow$ “some girls are girls that are dancing”